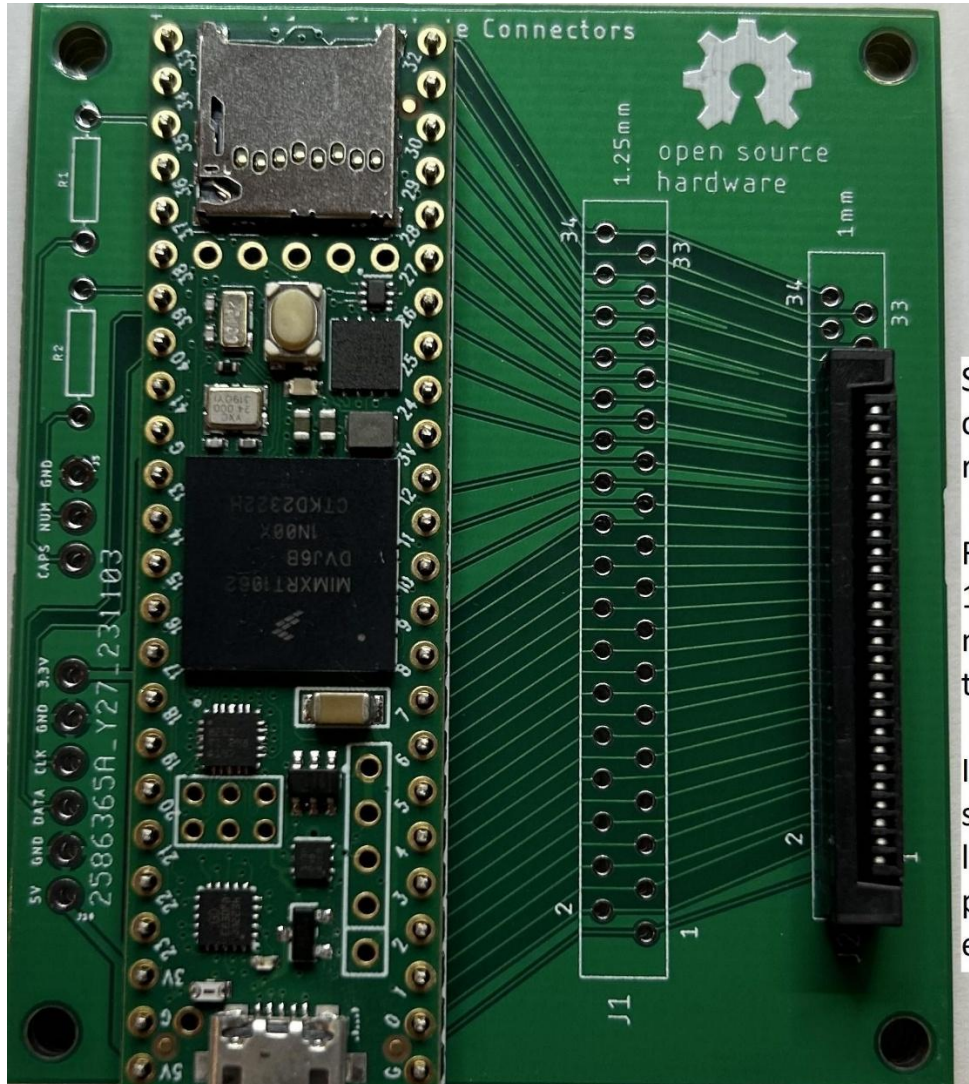


Teensy 4.1 "Easy-Soldering Board"

If your soldering skills are not ready for surface mount pads but you can solder a pin in a hole, the board below uses a vertical FPC connector with through hole pins.



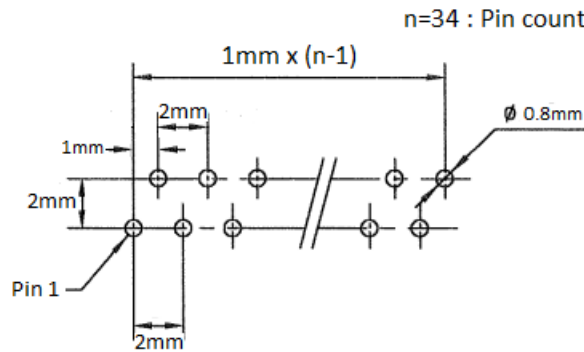
Shown with
contacts facing
right.

Rotate connector
180 degrees to
make contacts on
the left instead.

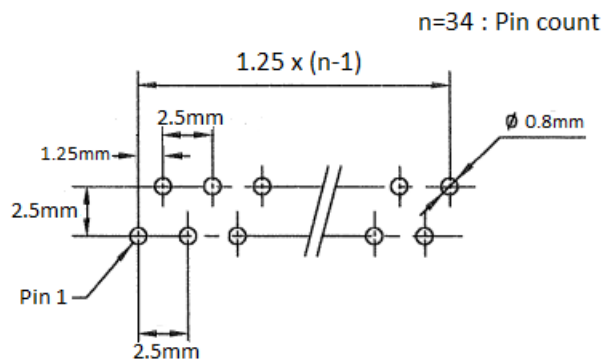
Insert connector
starting at pin 1
leaving unused
pins on the other
end.

This board only works if your keyboard FPC cable has no more than 34 pins and a pitch of 1mm or 1.25mm (see dimensions below).

Circuit Board Hole Pattern for 1mm pitch FPC connector



Circuit Board Hole Pattern for 1.25mm pitch FPC connector



Download the Eagle file "T4p1_thruhole.brd" or Gerber file "T4p1_thruhole_2023-11-01.zip" from my [repo](#) for fabrication. The board has 34 standard plated through holes for the connector pins instead of surface mount pads. You can rotate the connector 180 degrees so the contacts are on the left or the right to match up with your FPC cable. The pin spacing is designed for vertical connectors such as the 26 pin, 1mm pitch [Amphenol HLW26S-2C7LF](#). Replace the 26 in the part number with the number of pins in your cable. These or similar connectors are available from supply houses like [Digikey](#), [Mouser](#), [Arrow](#), [Newark](#), and [AliExpress](#). The 1.25mm pitch connectors (used on very old laptops) are a little harder to find but there are some available from [AliExpress](#).

If you want to drive CAPS or NUM Lock LEDs, there are pads for axial leaded current limit resistors driven by I/O's 37 and 38 (see schematic below). To control a 5 volt touchpad, I/O's 39 and 40 are brought to pads that can be wired to an [Adafruit 757](#) level translator.

